Harvey Williams | harveywilliams.com | hw562@cantab.ac.uk

github.com/harveywilliams56

Skills

- **Programming:** Github; Python; C++; Golang; Linux CLI
- **Mathematics:** Markov Chains; Measure Theory; Dynamical Systems; LaTeX
- **Machine Learning:** PyTorch; JAX; Monte Carlo Methods

Profile

My expertise is in physics-inspired algorithm design and verification, drawing on my two degrees in physics and pure mathematics — I have completed several internships and coursework projects analysing stochastic and deterministic algorithms, including machine learning methods.

I have exceptional determination and accountability, having worked part time during my A-levels and full time for two gap years to fund my undergraduate degree. I have also self-taught a substantial volume of pure mathematics and secured a Martingale Masters Scholarship.

In addition to the technical experience outlined below, I have taken on a variety of customer facing roles, including bar staff and site first aider.

Education

University of Manchester / Martingale Scholarship in Pure Mathematics — Sept 2025 - Sept 2026

- Fully funded 12 month MSc with both taught and research components
- Dissertation on the analysis of complex dynamical systems
- Semester 1: Measure and Ergodic Theory; Dynamical Systems; Manifolds; Riemann Surfaces
- Semester 2: Algebraic Topology; Lie Algebras

University of Cambridge / MSc Astrophysics — Oct 2020 - June 2025

- Specialism in theoretical physics, computational methods and machine learning
- Research project on statistical sampling in high dimensions — eg Nested Sampling
- Proved the correctness of a novel Hamiltonian Monte Carlo sampling algorithm variant
- Implemented my research project using JAX
- Part III Exams: Quantum Field Theory; Symmetries; General Relativity; Astrostatistics

The Blue School Sixth Form (Somerset UK) / A-level — Sept 2016 - July 2018

- Mathematics (**A***); Further Mathematics (**A***); Chemistry (**A***); Physics (**A***); Biology (**A**)

Experience

Freelance - Machine Learning / Summer School Leader — June 2025 - Aug 2025

- Introduced Loss Functions, Automatic Differentiation and Gradient Descent
- Demonstrated the influence of learning rate on convergence using linear regression
- Discussed how this generalises to neural nets and CNN architectures
- Demonstrated transfer learning with the Fast.AI library in Python

Gurdon Institute/Cambridge Centre for Physical Biology / Internship June 2024 - Aug 2024

- Wrote a computer simulation of collective cell motion based on research papers
- Sped up previous code by a factor of **20** whilst improving clarity by rewriting in Golang
- Analysed scaling of possible algorithms
- Proposed and implemented a more explainable approach (Delaunay Triangulation)
- Advised groups at Oxford and University of Chile carrying out similar research

Cambridge Centre for Carbon Credits / Internship June 2023 - Sept 2023

- Built an end-to-end ETL data cleaning pipeline from remote SQL server to labelled GIS file
- Used chunked processing to reduce peak memory consumption from **16GB** to **1GB**
- Wrote data fetching classes and subroutines for keeping SSH connection alive/reconnecting
- Strong security awareness using a local SSH key file and a config file to point to it
- Optimised processing speed by pre-processing using data smoothing
- Quantified **X%** suspected fraud and **Y%** uncertainty on CO2 capture — X and Y confidential
- Libraries: SQLAlchemy, Pandas, GeoPandas, Shapely, Numpy

Starleaf Cambridge / Year in Industry (YINI) Aug 2019 - Aug 2020

- Designed an ALU in a Hardware Description Language — Nand2Tetris course
- Wrote API based code in Python for a R&D project
- Automated visual testing with OpenCV-Python
- Presented test results to other members of the team
- Promptly flagged a major concern, preventing flawed units from being shipped to customers
- Organised a group entry into the Google Code Jam competition for the interns/grads

The Blue School / Teaching Assistant (maths dept.) Oct 2018 - April 2019

- Coordinated with teachers to support GCSE pupils in class
- Planned and delivered small group lessons

Achievements and Volunteering

- Organiser and webmaster for MIMUC 2026 mathematics conference / www.mimuc.org.uk
- Created a successful funding lead for MIMUC 2027 by networking at a careers fair
- Martingale Scholar 2025-2026
- Cambridge University Rifle Association — safe handling and use of firearms
- Selected as Vice-Captain for University Challenge 2025, Homerton College
- eCambassador at the Cambridge Admissions Office — mentoring disadvantaged applicants
- Part IB Physics Rep — successfully lobbied for the release of existing examples
- Rowed competitively for Homerton College Boat Club
- I'm also a keen runner and would love to join any office charity runs